Homework Solutions

Course Math 1A Worksheet 3 · **Student** Songpengfei · **Date** 2026-10-09 · **Solved** 8/8 (100.0%)

Problem Q1

Can the graph of a function have more than one tangent at a given point? If so, graph your answer. If not, explain why.

1. In general, a smooth function has exactly one tangent line at each point.
2. However, a curve (not necessarily the graph of a function) can have multiple tangents at a self-intersection point.
3. For a function graph, the tangent is unique where the derivative exists.

> **Answer** $\textbf{Unique tangent where } f'(a) \textbf{ exists.}$

`conceptual_template` calculus_limits

Problem Q2

Is there a function whose graph doesn't have a tangent at some point? If so, graph your answer. If not, explain why.

1. Yes. A function may fail to have a tangent at a point where it is not differentiable.
2. Examples: $f(x) = |x|$ at $x = 0$ (corner), $f(x) = x^{1/3}$ at $x = 0$ (vertical tangent/cusp).

> **Answer** $\textbf{Yes: } f(x) = |x| \textbf{ at } x = 0 \textbf{ (corner)}$

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Problem P1

Suppose δ is a positive real number. How do you describe all real numbers x that are within δ of 0?

1. x within δ of 0: $|x| < \delta$, i.e. $-\delta < x < \delta$

   (a) $-\delta < x < \delta$

   (b) $|x| < \delta$

   (c) $(-\delta, \delta)$

> **Answer** $|x| < \delta$, i.e. $-\delta < x < \delta$

`epsilon_delta_template` calculus_limits

Problem P2

Suppose a is a real number and δ is a positive real number. How do you describe all real numbers x that are within δ of a?

1. x within δ of a: $|x - a| < \delta$, i.e. $a - \delta < x < a + \delta$

   (a) $\mathrm{See\ graph}$

   (b) $a - \delta < x < a + \delta$

   (c) $|x - a| < \delta$

   (d) $(a - \delta, a + \delta)$

> **Answer** $|x - a| < \delta$, i.e. $a - \delta < x < a + \delta$

`epsilon_delta_template` calculus_limits

Problem P3

Suppose a is a real number and δ is a positive real number. How do you describe all real numbers x that are within δ of a, but not equal to a:

1. x within δ of a, not equal to a: $0 < |x - a| < \delta$

   (a) See graph

   (b) $a - \delta < x < a + \delta, \quad x \neq a$

   (c) $0 < |x - a| < \delta$

   (d) $(a - \delta, a) \cup (a, a + \delta)$

> **Answer** $0 < |x - a| < \delta$

`epsilon_delta_template` calculus_limits

Problem P4

Let $f(x) = x^2$.

1. Given: $f(x) = x^2$

   (a) $(2.82842712474619, 3.16227766016838)$

   (b) $\delta = 0.16227766016838$

   > **(c)** 未解：无法解析 ε-δ 计算子题

   (d) Yes (continuity)

> **Answer** $f(x) = x^2$

`epsilon_delta_computation` calculus_limits

Problem AP1

Graph $y = 2x - x^2$. For which points a is the tangent line to the curve at the point $(a, 2a - a^2)$ a horizontal line? How is the value of $2x - x^2$ at these points related to the values of $2x - x^2$ at other points?

1. Given: $y = -x^2 + 2x$
2. Derivative: $y' = 2 - 2x$
3. Horizontal tangent where $y' = 0$: $2 - 2x = 0$
4. Solutions: $x = [1]$
5. Points: $(1, 1)$
6. $f''(1) = -2 < 0$: this is a maximum.
7. The value 1 is the maximum of $-x^2 + 2x$.

> **Answer** $(1, 1)$

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Problem AP2

The point $(2, 0)$ is on the curve $y = 2x - x^2$.

1. Given: $y = -x^2 + 2x$
2. Derivative: $y' = 2 - 2x$
3. At point $(2, 0)$:
4. Slope: $f'(2) = -2$
5. Tangent line: $y = 4 - 2x$

   (a) $\text{Line with slope } -2$

   (b) $y = 4 - 2x$

> **Answer** $4 - 2x$

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